SSC MTS 2024 question paper with answers

1 October 2024 · Shift 2

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

If 4% of (A + B) is equal to 28% of (A – B), then find A : B.

(A) 3 : 4
(B) 4 : 3
(C) 2 : 3
(D) 1 : 3

Q.2

By selling an article for ₹19, a dealer makes a profit of 20%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 40%?

(A) 3.17
(B) 5.17
(C) 4.17
(D) 6.17

Q.4

A 725 m long train passes through a 235 m long tunnel in 48 sec. Find the speed of the train.

(A) 82 km/h
(B) 54 km/h
(C) 66 km/h
(D) 72 km/h

Q.5

A train travelling at a speed of 60 km/h leaves station A at 7:00 a.m. and another train travelling at a speed of 84 km/h starts from station A at 9:30 a.m. in the same direction on a parallel track. At what distance (in km) from station A and after how many hours of travel of the second train will they be together? Express your answer in the same order.

(A) 375, 6.25
(B) 693, 8.25
(C) 525, 6.25
(D) 495, 8.25

Q.6

The average weight of a group of 20 women was calculated as 58.5 kg and it was later discovered that the weight of one of the women was read as 35 kg, whereas her actual weight was 65 kg. What is the actual average weight of the group of 20 women?

(A) 40 kg
(B) 50 kg
(C) 70 kg
(D) 60 kg

Q.8

A merchant has 5000 kg of Basmati rice. He sells some part of this Basmati rice at 20% profit and the rest at 35% profit. He gains 29% in the entire transaction. What is the quantity sold at 20% profit and 35% profit, respectively?

(A) 2600 kg, 2400 kg
(B) 2200 kg, 2800 kg
(C) 2000 kg, 3000 kg
(D) 1800 kg, 3200 kg

Q.9

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 4% per year rate of interest for 4 years.

(A) 440
(B) 400
(C) 360
(D) 340

Q.11

A shopkeeper makes a 15% profit on a sale. He would have made 18%, if he had sold it for Rs.18 more. The cost price of the article is:

(A) Rs.600
(B) Rs.500
(C) Rs.550
(D) Rs.650

Q.12

Find the HCF of 24, 36 and 60.

(A) 4
(B) 6
(C) 12
(D) 360

Q.13

Sunil, Basid, Vikram, and Kuldeep receive salaries in the ratio of 3 : 7 : 8 : 13. If the difference between the salaries of Sunil and Basid is ₹2,000, then what is the difference between the salaries (in ₹) of Vikram and Kuldeep?

(A) 2,400
(B) 2,200
(C) 2,000
(D) 2,500

Q.14

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 4% per year rate of interest for 5 years.

(A) 460
(B) 540
(C) 440
(D) 500

Q.15

A class of 30 students appeared in a test. The average score of 12 students is 80, and that of the rest is 70. What is the average score of the class?

(A) 87
(B) 77
(C) 74
(D) 67

Q.16

If tea that costs ₹7.50 per kg is mixed with tea that costs ₹10.50 per kg in the proportion of 2 :1 then, what is the cost (in ₹) of the resultant mixture per kg?

(A) 7.50
(B) 8.50
(C) 9.00
(D) 7.00

Q.17

Manpreet offers two successive discounts of 12% and 8% on the sale of a sweater to his customers on the occasion of Diwali. What is the equivalent single percentage discount offered by Manpreet?

(A) 19.64%
(B) 20.96%
(C) 19.04%
(D) 20.00%

Q.18

A person deposited ₹2,500 in a bank that pays 8% simple interest. He withdrew ₹1,000 (from amount) at the end of the first year. What will be his balance after 4 years?

(A) ₹2,018
(B) ₹2,108
(C) ₹2,280
(D) ₹2,480

Reasoning Ability and Problem Solving 16 questions

Q.1

10 is related to 75 following a certain logic. Following the same logic, 6 is related to 47. To which of the following is 15 related following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 110
(B) 156
(C) 123
(D) 164

Q.2

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 456 × 4 – 3 ÷ 11 + 70 =?

(A) 77
(B) 87
(C) 79
(D) 84

Q.4

MJHI is related to TQOP in a certain way. In the same way, GDBC is related to NKIJ. To which of the following is XUST related following the same logic?

(A) BZAC
(B) EABZ
(C) EBZA
(D) DABZ

Q.6

What will come in the place of '?' in the following equation, if '+' and '×' are interchanged and '–' and '÷' are interchanged? 17 + 77 – 7 × 37 ÷ 47 = ?

(A) 177
(B) 157
(C) 187
(D) 167

Q.7

In a certain code language, 'VOLT' is coded as '3517' and 'TURN' is coded as '2348'. What is the code for 'T' in that language?

(A) 3
(B) 2
(C) 7
(D) 8

Q.8

Seven people R, S, T, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). T is sitting to the immediate right of G. R is sitting to the immediate right of E. Only two people are sitting between D and S when counted from the right of S. E is an immediate neighbour of S. If F is not an immediate neighbour of S, who is sitting third to the left of F?

(A) T
(B) E
(C) S
(D) R

Q.10

Six friends are sitting on a circular table, facing towards the centre (but not necessarily in the same order) . X is to the immediate left of Z and W is to the immediate right of Y. Z and U are separated by only one person sitting between them. W is an immediate neighbour of U. Only V is sitting between Z and Y. Who among the following are the immediate neighbours of U?

(A) W and V
(B) W and Y
(C) X and W
(D) Z and Y

Q.11

In a certain code language, ‘make some noise’ is written as ‘ja sp ko’ and ‘noise is painful’ is writtenas ‘la ma sp’. How will ‘noise’ be writtenin that language?

(A) la
(B) ma
(C) sp
(D) ko

Q.12

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 8, 30; 12, 46

(A) 16, 62
(B) 15, 61
(C) 15, 62
(D) 16, 61

Q.13

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All shirts are trousers. No trouser is a belt. Conclusions: (I) No shirt is a belt. (II) Some belts may be trousers.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Only conclusion II follows
(D) Both conclusions I and II follow

Q.14

In a certain code language, 'BORDER' is coded as ‘OBDRRE’ and 'BEYOND' is coded as ‘EBOYDN’. How will 'BURDEN' be coded in the same language?

(A) UBDREN
(B) NEDRUB
(C) UBDRNE
(D) UBRDNE

Q.15

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: Some whales are sharks. No shark is a dolphin. Conclusions: (I) No dolphin is a shark. (II) Some sharks are whales.

(A) Both conclusions I and II follow.
(B) Only conclusion I follows.
(C) Neither conclusion I nor II follows.
(D) Only conclusion II follows.

Q.17

Which of the following letter-clusters will replace the question mark (?) in the given series to make it logically complete? EMU, DNT, COS, BPR, ?

(A) BST
(B) FCD
(C) AQQ
(D) UBO

Q.18

What should come in place of the question mark (?) in the given series? 10, 22, 46, 94, 190, ?

(A) 385
(B) 389
(C) 382
(D) 387

Q.19

CE 77 is related to IK 87 in a certain way. In the same way, GJ 52 is related to MP 62. To which of the following is BG 43 related following the same logic?

(A) HM 53
(B) NZ 30
(C) XY 35
(D) WZ 20

Q.20

In a certain code language, 'BRIGHT' is coded as '371562', and 'GIRTHS' is coded as '653782'. What is the code for ‘B’ in 'BRIGHT'?

(A) 7
(B) 8
(C) 2
(D) 1

General Awareness 24 questions

Q.1

The autobiography of which of the following actors is titled ‘The Substance And The Shadow: An Autobiography’?

(A) Rishi Kapoor
(B) Rajesh Khanna
(C) Dilip Kumar
(D) Vinod Khanna

Q.2

Which Indian dancer of the Manipuri style won the Tagore Award for Cultural Harmony in 2014?

(A) Elam Endira Devi
(B) Rajkumar Singhajit Singh
(C) Nirmala Mehta
(D) Guru Bipin Sinha

Q.3

Which edition of the Intercontinental Cup event was played in India in 2023?

(A) 5th
(B) 10th
(C) 1st
(D) 3rd

Q.4

The Steel Authority of India Limited (SAIL) is an example of the _____ industry.

(A) joint sector
(B) private sector
(C) public sector
(D) cooperative sector

Q.5

Which classical dance form is performed by the renowned artists; Sitara Devi, Rohini Bhate, Pandit Munnalal Shukla, Kumudini Lakhia and Uma Sharma?

(A) Kathak
(B) Bharatanatyam
(C) Odissi
(D) Kuchipudi

Q.6

When was the All Bengal Students Conference held in the chairmanship of Jawaharlal Nehru?

(A) 1939
(B) 1920
(C) 1934
(D) 1928

Q.7

In the context of microfinance, what does ‘S’ stand for in SHG?

(A) Steady
(B) Self
(C) Sustainable
(D) Social

Q.8

As per Article 51A of the Constitution of India, Fundamental Duty (h) is to develop the _____, humanism and the spirit of inquiry and reform.

(A) scientific temper
(B) sociological temper
(C) religious temper
(D) cultural temper

Q.9

What is the name of Formula E's first ever race in India, which was hosted in collaboration with the Telangana government?

(A) The 2023 Greenko Hyderabad E-Prix
(B) The 2023 Greenko Buddha Circuit E-Prix
(C) The 2023 Mahindra Bangalore E-Prix
(D) The 2023 Mahindra Mumbai E-Prix

Q.10

Which of the following statements is/are FALSE for the Harappan town planning? 1. Most of the towns have citadel in harappan civilization. 2. Every town had a temple in Harappan civilization. 3. Every building was made of sand stone in Harappan civilization.

(A) Only 1
(B) Both 2 and 3
(C) Both 1 and 3
(D) Both 1 and 2

Q.11

The Ramoshi Uprising of 1824 took place in which of the following present states of India?

(A) Tamil Nadu
(B) Uttar Pradesh
(C) West Bengal
(D) Maharashtra

Q.13

In October 2023, Maharashtra government planned which of the following schemes to overhaul its medical system after the death of 38 patients in Nanded?

(A) Ayushman Maharashtra
(B) Arogya Mission
(C) Sahakar Mitra
(D) Vision 2035

Q.14

A process like river bifurcation can create valuable landforms for the environment, providing growth of vegetation. An example of such landform is _____.

(A) cascades
(B) river deltas
(C) potholes
(D) V-shaped valleys

Q.15

According to the National Multidimensional Poverty Index 2021, which Union Territory has the lowest proportion of poor?

(A) Chandigarh
(B) Puducherry
(C) Lakshadweep
(D) Delhi

Q.16

The Taj Mahal was built by who among the following?

(A) Shah Jahan
(B) Akbar
(C) Babar
(D) Jahangir

Q.17

During which centuries did several Buddhist pilgrims and scholars travel to China on the historic 'silk route'?

(A) Eighth and ninth centuries AD
(B) Sixth and seventh centuries AD
(C) First, second and third centuries AD
(D) Fourth and fifth centuries AD

Q.18
Which Article of the Constitution of India specifically addresses the qualifications for the office of the President?

(A) Article 60
(B) Article 58
(C) Article 50
(D) Article 56

Q.19
Non-polar molecules possess _____.

(A) partial charge
(B) instantaneous dipole moment
(C) no dipole moment
(D) permanent dipole moment

Q.20
In which state is Kawal Wildlife Sanctuary located?

(A) Maharashtra
(B) Gujarat
(C) Odisha
(D) Telangana

Q.21
Who among the following was the most important god of the Aryans?

(A) Marut
(B) Pashupatinath
(C) Indra
(D) Agni

Q.22
Who among the following led the Parliamentary delegation to participate in the 9th BRICS Parliamentary Forum in September 2023?

(A) Nitin Gadkari
(B) Jagdeep Dhankar
(C) Harivansh Narayan Singh
(D) Om Birla

Q.23
_____ was the pioneer who modified the Hawaiian slide guitar to play classical ragas.

(A) Buddhadev Dasgupta
(B) Brij Bhushan Kabra
(C) Palghat Raghu
(D) Krishna Ram Chaudhary

Q.24
Select the synthetic indicator from among the given options.

(A) Turmeric
(B) Methyl orange
(C) Litmus solution
(D) Cabbage juice

Q.25
In which state is 'Bhogali Bihu' festival celebrated?

(A) Jharkhand
(B) Chhattisgarh
(C) Assam
(D) Gujarat

English Language and Comprehension 22 questions

Q.1
Select the word that has the same meaning (SYNONYM) as the underlined word in the given sentence. She accused her sister of being “vindictive” when she was trying to defame her.

(A) revengeful
(B) venerable
(C) vigilant
(D) righteous

Q.2

Select the most appropriate meaning of the given idiom. When pigs fly

(A) Something that spoils relationships
(B) Something that will never happen
(C) Something terrible
(D) Something unplanned

Q.3

Select the INCORRECTLY spelt word from the given sentence. The artist's unique technique added a touch of individiuality to the painting.

(A) Unique
(B) Artist's
(C) Technique
(D) Individiuality

Q.4

Select the most appropriate meaning of the given idiom. Apple of one's eye

(A) Very precious
(B) Very docile
(C) Very stubborn
(D) Very strong

Q.5

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. The children were “merry” than usual on Sunday.

(A) merrier
(B) more merrier
(C) merriest
(D) the merriest

Q.6

Select the most appropriate synonym of the given word. Blunt

(A) Rowdy
(B) Adventurous
(C) tactful
(D) Insensitive

Q.7

Change the following sentence into superlative degree. No other mountain in the world is as high as the Mount Everest.

(A) No other mountain is highest than Mount Everest.
(B) Mount Everest is the highest mountain in the world.
(C) Mount Everest is higher than any other mountain in the world.
(D) Mount Everest is high mountain in the world.

Q.8

Select the sentence with the most appropriate use of preposition.

(A) Reetu slept all through the history class.
(B) Reetu slept all during the history class.
(C) Reetu slept all into the history class.
(D) Reetu slept all in the history class.

Q.9

Select the most appropriate ANTONYM of the underlined word in the given sentence. The children's laughter was filled with “exuberant” joy.

(A) sarcastic
(B) repressed
(C) euphoric
(D) flawless

Q.10

Read the sentence given below and identify if there is an error in it. The error can be in any segment of the sentence which are marked A, B, C, and D. Select option D if there is 'No error'. (A) Ritika angrily / (B) spoke to Kartik / (C) after the meeting / (D) No error.

(A) B
(B) C
(C) A
(D) D

Q.11

Select the option that will improve the underlined part of the given sentence. In case no improvement is needed, select 'No improvement required'. Placing a huge magnet atop the temple “pose to being” the biggest hurdle for the masons.

(A) poses for being
(B) posed to be
(C) posing as be
(D) No improvement required

Q.12

Select the most appropriate option to fill in the blank. My grandparents live in a beautiful house _____ the countryside.

(A) at
(B) on
(C) in
(D) onto

Q.14

Select the most appropriate option to substitute the underlined segment in the given sentence. They “looked out the matter and found” the root cause of the problem.

(A) looked until the matter and found
(B) looked into the matter and found
(C) looked unless the matter and found
(D) looked of the matter and found

Q.15

Parts of the following sentence have been given as options. Select the option that contains an error. The farm cattle is grazing in the field when the tiger attacked.

(A) in the field when
(B) The farm cattle
(C) is grazing
(D) the tiger attacked.

Q.17

Select the most appropriate ANTONYM of the given word. Enmity

(A) Friendship
(B) Hostility
(C) Malignity
(D) Antipathy

Q.18

Select the most appropriate option to fill in the blank. The loud thunder _____ the children, making them cry.

(A) fainted
(B) frightened
(C) fancied
(D) fastened

Q.19

Select the option that expresses the given sentence in future tense form. Since next Friday, the team of three members has been tracking the animal.

(A) From next Friday, the team of three members have been tracking the animal.
(B) From next Friday, the team of three members are tracking the animal.
(C) From next Friday, the team of three members will be tracking the animal.
(D) Since last Friday, the team of three members had tracked the animal.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A monsoon is traditionally a seasonal reversing wind (1)_____ by corresponding changes in precipitation but is now (2)_____ describe seasonal changes in atmospheric circulation and precipitation associated (3)_____ annual latitudinal oscillation of the Intertropical Convergence Zone (ITCZ) between its limits to the north and south of the equator. Usually, the term monsoon (4)_____ refer to the rainy phase of a seasonally changing pattern, although technically there is also a dry phase. The term is also sometimes used to (5)_____ locally heavy but short-term rains.

Select the most appropriate option to fill in blank number 1.

(A) accompanies
(B) accompanying
(C) accompanied
(D) accompany

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A monsoon is traditionally a seasonal reversing wind (1)_____ by corresponding changes in precipitation but is now (2)_____ describe seasonal changes in atmospheric circulation and precipitation associated (3)_____ annual latitudinal oscillation of the Intertropical Convergence Zone (ITCZ) between its limits to the north and south of the equator. Usually, the term monsoon (4)_____ refer to the rainy phase of a seasonally changing pattern, although technically there is also a dry phase. The term is also sometimes used to (5)_____ locally heavy but short-term rains.

Select the most appropriate option to fill in blank number 2.

(A) uses to
(B) use to
(C) used to
(D) using to

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A monsoon is traditionally a seasonal reversing wind (1)_____ by corresponding changes in precipitation but is now (2)_____ describe seasonal changes in atmospheric circulation and precipitation associated (3)_____ annual latitudinal oscillation of the Intertropical Convergence Zone (ITCZ) between its limits to the north and south of the equator. Usually, the term monsoon (4)_____ refer to the rainy phase of a seasonally changing pattern, although technically there is also a dry phase. The term is also sometimes used to (5)_____ locally heavy but short-term rains.

Select the most appropriate option to fill in blank number 3.

(A) with
(B) on
(C) of
(D) for

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A monsoon is traditionally a seasonal reversing wind (1)_____ by corresponding changes in precipitation but is now (2)_____ describe seasonal changes in atmospheric circulation and precipitation associated (3)_____ annual latitudinal oscillation of the Intertropical Convergence Zone (ITCZ) between its limits to the north and south of the equator. Usually, the term monsoon (4)_____ refer to the rainy phase of a seasonally changing pattern, although technically there is also a dry phase. The term is also sometimes used to (5)_____ locally heavy but short-term rains.

Select the most appropriate option to fill in blank number 4.

(A) is used to
(B) is use to
(C) are used to
(D) were used to

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A monsoon is traditionally a seasonal reversing wind (1)_____ by corresponding changes in precipitation but is now (2)_____ describe seasonal changes in atmospheric circulation and precipitation associated (3)_____ annual latitudinal oscillation of the Intertropical Convergence Zone (ITCZ) between its limits to the north and south of the equator. Usually, the term monsoon (4)_____ refer to the rainy phase of a seasonally changing pattern, although technically there is also a dry phase. The term is also sometimes used to (5)_____ locally heavy but short-term rains.

Select the most appropriate option to fill in blank number 5.

(A) describe
(B) describes
(C) described
(D) describing

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(A)**	Q.4 **(D)**	Q.5 **(C)**	Q.6 **(D)**	Q.8 **(C)**
Q.9 **(B)**	Q.11 **(A)**	Q.12 **(C)**	Q.13 **(D)**	Q.14 **(D)**	Q.15 **(C)**
Q.16 **(B)**	Q.17 **(C)**	Q.18 **(B)**			

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(A)**	Q.4 **(C)**	Q.6 **(A)**	Q.7 **(A)**	Q.8 **(B)**
Q.10 **(C)**	Q.11 **(C)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(C)**	Q.15 **(A)**
Q.17 **(C)**	Q.18 **(C)**	Q.19 **(A)**	Q.20 **(D)**		

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(D)**	Q.13 **(D)**
Q.14 **(B)**	Q.15 **(B)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(B)**	Q.19 **(B)**
Q.20 **(D)**	Q.21 **(C)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(B)**	Q.25 **(C)**

English Language and Comprehension

Q.1 **(A)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(A)**	Q.5 **(A)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(B)**	Q.12 **(C)**
Q.14 **(B)**	Q.15 **(C)**	Q.17 **(A)**	Q.18 **(B)**	Q.19 **(C)**	Q.21 **(C)**
Q.22 **(C)**	Q.23 **(A)**	Q.24 **(A)**	Q.25 **(A)**		

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 4 : 3

4(A + B) = 28(A – B), so A + B = 7A – 7B and 6A = 8B. A : B = 8 : 6 = 4 : 3.

Q.2 · Answer (A) 3.17

Cost price = 19 ÷ 1.2 = ₹15.833. At 40% profit the price is 15.833 × 1.4 = ₹22.167, an increase of 22.167 – 19 = ₹3.17.

Q.4 · Answer (D) 72 km/h

The train covers its own length plus the tunnel's, 725 + 235 = 960 m, in 48 s: 20 m/s = 20 × 18/5 = 72 km/h.

Q.5 · Answer (C) 525, 6.25

By 9:30 a.m. the first train is 2.5 × 60 = 150 km ahead. The second gains 84 – 60 = 24 km/h, so it draws level after 150/24 = 6.25 hours, 84 × 6.25 = 525 km from A.

Q.6 · Answer (D) 60 kg

The total was 65 – 35 = 30 kg too low, so the average rises by 30/20 = 1.5 kg: 58.5 + 1.5 = 60 kg.

Q.8 · Answer (C) 2000 kg, 3000 kg

By alligation the quantities sold at 20% and 35% are in the ratio (35 – 29) : (29 – 20) = 6 : 9 = 2 : 3: 2,000 kg and 3,000 kg of the 5,000 kg.

Q.9 · Answer (B) 400

SI = 2,500 × 4 × 4/100 = ₹400.

Q.11 · Answer (A) Rs.600

The extra ₹18 is 18% – 15% = 3% of the cost price, so the cost price is 18 × 100/3 = ₹600.

Q.12 · Answer (C) 12

24 = $2^3 \times 3$, 36 = $2^2 \times 3^2$ and 60 = $2^2 \times 3 \times 5$. The factors common to all three give $2^2 \times 3 = 12$.

Q.13 · Answer (D) 2,500

Basid's 7 parts minus Sunil's 3 is 4 parts = ₹2,000, so one part is ₹500. Kuldeep's 13 minus Vikram's 8 is 5 parts = ₹2,500.

Q.14 · Answer (D) 500

SI = 2,500 × 4 × 5/100 = ₹500.

Q.15 · Answer (C) 74

(12 × 80 + 18 × 70)/30 = (960 + 1,260)/30 = 2,220/30 = 74.

Q.16 · Answer (B) 8.50

(2 × 7.50 + 1 × 10.50)/3 = 25.50/3 = ₹8.50 a kg.

Q.17 · Answer (C) 19.04%

Successive discounts of a% and b% equal a + b – ab/100: 12 + 8 – 0.96 = 19.04%.

Q.18 · Answer (B) ₹2,108

After the first year the balance is 2,500 + 200 = ₹2,700, and ₹1,700 after the withdrawal. Three more years at 8% add 1,700 × 8 × 3/100 = ₹408: ₹2,108.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) 110

Each number is multiplied by 7 and 5 is added: 10 × 7 + 5 = 75 and 6 × 7 + 5 = 47. Likewise 15 × 7 + 5 = 110.

Q.2 · Answer (A) 77

After interchanging the signs: 456 ÷ 4 + 3 × 11 – 70 = 114 + 33 – 70 = 77.

Q.4 · Answer (C) EBZA

Each letter moves 7 places forward: M → T, J → Q, H → O, I → P. So X → E, U → B, S → Z and T → A (wrapping round after Z): EBZA.

Q.6 · Answer (A) 177

After interchanging the signs: 17 × 77 ÷ 7 + 37 − 47 = 17 × 11 + 37 − 47 = 187 + 37 − 47 = 177.

Q.7 · Answer (A) 3

T is the only letter in both words and 3 the only digit in both codes, so T = 3.

Q.8 · Answer (B) E

Clockwise round the table: S, T, G, F, D, R, E (facing the centre, each person's left is the next seat clockwise). E cannot sit to S's left, as R, to E's right, would then take S's seat; and F, kept away from S, sits beside D. The three seats to F's left are D, R and E.

Q.10 · Answer (C) X and W

Clockwise round the table: Z, X, U, W, Y, V (facing the centre, each person's left is the next seat clockwise). X is on Z's left, so V and then Y are on Z's right; W sits to Y's right, and U, next to W, is two seats from Z. U's neighbours are X and W.

Q.11 · Answer (C) sp

'noise' is the only word in both sentences and 'sp' the only code in both, so noise = sp.

Q.12 · Answer (A) 16, 62

The second number is four times the first minus 2: 8 × 4 − 2 = 30 and 12 × 4 − 2 = 46. Likewise 16 × 4 − 2 = 62.

Q.13 · Answer (A) Only conclusion I follows

All shirts are trousers and no trouser is a belt, so no shirt is a belt (I). II cannot follow: as no trouser is a belt, no belt can be a trouser.

Q.14 · Answer (C) UBDRNE

The letters of each pair are swapped: BO RD ER → OB DR RE and BE YO ND → EB OY DN. So BU RD EN → UB DR NE: UBDRNE.

Q.15 · Answer (A) Both conclusions I and II follow.

'No shark is a dolphin' converts to 'no dolphin is a shark' (I), and 'some whales are sharks' to 'some sharks are whales' (II). Both follow.

Q.17 · Answer (C) AQQ

First letters move back one (E, D, C, B, A), second letters forward one (M, N, O, P, Q) and third letters back one (U, T, S, R, Q): AQQ.

Q.18 · Answer (C) 382

Each term is twice the one before plus 2: 10 × 2 + 2 = 22, 22 × 2 + 2 = 46, and so on to 190 × 2 + 2 = 382.

Q.19 · Answer (A) HM 53

Each letter moves 6 places forward and the number rises by 10: C → I, E → K, 77 → 87. So B → H, G → M and 43 → 53: HM 53.

Q.20 · Answer (D) 1

R, I, G, H and T are in both words, and 3, 7, 5, 6 and 2 in both codes. The letter left in BRIGHT is B and the digit left in its code is 1, so B = 1.

General Awareness

Q.1 · Answer (C) Dilip Kumar

'The Substance and the Shadow', published in 2014, is the autobiography of the actor Dilip Kumar.

Q.2 · Answer (B) Rajkumar Singhajit Singh
Rajkumar Singhajit Singh, a senior guru of Manipuri dance, received the Tagore Award for Cultural Harmony for 2014 (presented in 2019).

Q.3 · Answer (D) 3rd
The 2023 Intercontinental Cup, held in Bhubaneswar, was the football tournament's third edition, after those of 2018 and 2019; India won it.

Q.4 · Answer (C) public sector
SAIL is owned and run by the government, which makes it a public sector industry; a joint sector one is owned jointly by the state and private firms.

Q.5 · Answer (A) Kathak
Sitara Devi, Rohini Bhate, Munnalal Shukla, Kumudini Lakhia and Uma Sharma are all exponents of Kathak.

Q.6 · Answer (D) 1928
Jawaharlal Nehru presided over the All Bengal Students' Conference in Calcutta in 1928, which set up the All Bengal Students' Association.

Q.7 · Answer (B) Self
SHG stands for Self-Help Group: a small group, often of women, that pools its savings and lends to its members.

Q.8 · Answer (A) scientific temper
Article 51A(h) asks every citizen to develop the scientific temper, humanism and the spirit of inquiry and reform.

Q.9 · Answer (A) The 2023 Greenko Hyderabad E-Prix
India's first Formula E race was the Hyderabad E-Prix of February 2023, run on a street circuit in Hyderabad with the Telangana government.

Q.10 · Answer (B) Both 2 and 3
Statements 2 and 3 are false: no temples have been found in Harappan towns, and their buildings were mostly of baked brick, not sandstone. Many towns did have a raised citadel, so 1 is true.

Q.11 · Answer (D) Maharashtra
The Ramoshis, a community of the Western Ghats, rose against British rule in the 1820s around Satara and Pune, in present-day Maharashtra.

Q.13 · Answer (D) Vision 2035
In October 2023, after the patient deaths at the government hospital in Nanded, Chief Minister Eknath Shinde announced a 'New Healthcare Vision 2035' to overhaul the state's public health system.

Q.14 · Answer (B) river deltas
Near its mouth a river bifurcates into distributaries that spread silt into a delta, fertile land that supports rich vegetation.

Q.15 · Answer (B) Puducherry
In NITI Aayog's National Multidimensional Poverty Index 2021 (baseline report), Puducherry had the lowest share of poor people among the Union Territories, 1.72%.

Q.16 · Answer (A) Shah Jahan
The Mughal emperor Shah Jahan built the Taj Mahal at Agra in memory of his wife Mumtaz Mahal.

Q.17 · Answer (C) First, second and third centuries AD
SSC's key gives the first to third centuries AD, when Buddhism first reached China along the Silk Route and monks began carrying its texts there. (Pilgrims went on travelling the route in later centuries too.)

Q.18 · Answer (B) Article 58
Article 58 sets the qualifications for election as President; Article 56 deals with the term of office and Article 60 with the oath.

Q.19 · Answer (B) instantaneous dipole moment
SSC's key gives an instantaneous dipole moment: a non-polar molecule has no permanent dipole, but the momentary shifting of its electrons creates short-lived dipoles, the source of London dispersion forces. ('No dipole moment' is also arguable, as a description of its net, permanent state.)

Q.20 · Answer (D) Telangana
Kawal Wildlife Sanctuary, also a tiger reserve, is in northern Telangana.

Q.21 · Answer (C) Indra
Indra, god of rain and war, is the most important god of the Rig Veda, with about a quarter of its hymns addressed to him.

Q.22 · Answer (C) Harivansh Narayan Singh
Harivansh Narayan Singh, Deputy Chairman of the Rajya Sabha, led India's delegation to the 9th BRICS Parliamentary Forum in Johannesburg in September 2023.

Q.23 · Answer (B) Brij Bhushan Kabra
Brij Bhushan Kabra was the first to adapt the Hawaiian slide guitar to play Hindustani classical ragas.

Q.24 · Answer (B) Methyl orange
Methyl orange is made in the laboratory; turmeric, litmus (from lichens) and red cabbage juice are natural indicators.

Q.25 · Answer (C) Assam
Bhogali or Magh Bihu, the harvest festival held in January, is celebrated in Assam.

English Language and Comprehension

Q.1 · Answer (A) revengeful
'Vindictive' means wanting revenge: 'revengeful'.

Q.2 · Answer (B) Something that will never happen
'When pigs fly' describes something that will never happen.

Q.3 · Answer (D) Individiuality
The correct spelling is 'individuality'.

Q.4 · Answer (A) Very precious
The 'apple of one's eye' is a person or thing one holds very precious.

Q.5 · Answer (A) merrier
'Than' needs the comparative: 'merrier than usual'. 'More merrier' doubles the comparative.

Q.6 · Answer (D) Insensitive
A 'blunt' person speaks plainly, without regard for others' feelings: the closest option is 'insensitive'. 'Tactful' is its opposite.

Q.7 · Answer (B) Mount Everest is the highest mountain in the world.
'No other mountain is as high as Mount Everest' becomes, in the superlative degree, 'Mount Everest is the highest mountain in the world'.

Q.8 · Answer (A) Reetu slept all through the history class.
'All through' means from beginning to end: Reetu slept throughout the class.

Q.9 · Answer (B) repressed
'Exuberant' means full of unrestrained energy and cheer; its opposite is 'repressed', held back.

Q.10 · Answer (D) D
'Ritika angrily spoke to Kartik after the meeting' is correct as it stands.

Q.11 · Answer (B) posed to be
SSC's key gives 'posed to be': the sentence needs a finite verb in the past ('Placing a huge magnet atop the temple posed to be the biggest hurdle'), and of the four options only it supplies one. 'Pose to being', 'poses for being' and 'posing as be' are ungrammatical; in everyday English 'proved to be' would be the natural phrase.

Q.12 · Answer (C) in
One lives 'in' the countryside.

Q.14 · Answer (B) looked into the matter and found
To 'look into' a matter is to investigate it.

Q.15 · Answer (C) is grazing
'Cattle' is plural, and the attack was in the past: 'The farm cattle were grazing in the field when the tiger attacked.'

Q.17 · Answer (A) Friendship
'Enmity' is hostility; its opposite is 'friendship'.

Q.18 · Answer (B) frightened
Thunder that makes children cry has 'frightened' them.

Q.19 · Answer (C) From next Friday, the team of three members will be tracking the animal.
The future continuous 'will be tracking' puts the action in the future, with 'from next Friday' marking when it begins.

Q.21 · Answer (C) accompanied
A wind 'accompanied by' changes in precipitation: the past participle, used in a passive sense.

Q.22 · Answer (C) used to
The term 'is now used to describe' wider changes: 'used to' plus the verb, after 'is'.

Q.23 · Answer (A) with
Changes 'associated with' the annual oscillation of the ITCZ.

Q.24 · Answer (A) is used to
The singular subject 'the term monsoon' takes 'is used to refer to'.

Q.25 · Answer (A) describe
'Used to' is followed by the base form of the verb: 'used to describe'.

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